ADITYA THAKARE

AI Engineer | LangGraph Agents · RAG · FastAPI · MLOps

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SUMMARY

AI Engineer who builds agent backends that run in production. On contract for Neon AI, took over a legacy LangGraph codebase and shipped the agentic backend of an executive finance dashboard: **async FastAPI**, **DuckDB over Parquet**, MongoDB, and **Pydantic-validated** GPT-4o-mini output on every LLM call. Open source: **merged PR** in caura-ai's platform for long-running agents and two bug-fix PRs open on langchain-ai/langchain-mongodb. Benchmarked five router options for an agent workflow: a 5 MB XGBoost model matched a 1.6 GB fine-tuned transformer, and both beat a Gemini router by 20+ points.

PROFESSIONAL EXPERIENCE

AI Consultant & Developer (Contract) | **Neon AI** *May 2026 – Jul 2026*

- Built the agentic backend of a **production** executive analytics dashboard: a **LangGraph** multi-agent pipeline behind **async FastAPI** that answers natural-language financial questions.
- Took over a legacy LangGraph codebase and migrated it live with a **strangler-fig** approach, serving modern REST endpoints alongside the legacy ones so the frontend kept working throughout.
- Grounded every answer in data: **DuckDB over Parquet** for evidence queries, async **MongoDB** (Motor) for metadata, and **Pydantic** structured output on every GPT-4o-mini call through Azure OpenAI.
- Built a monitoring and root-cause-analysis agent system; handled chain concurrency with **asyncio** locks in a constrained single-node environment; evaluated cheaper models to replace GPT-4o-mini.
- Architecture comprises **8 graph nodes**, exposes **4 primary endpoints**, and handles average responses in **~4 seconds** for simple queries and **~12 seconds** for deep analytical queries.

Data Scientist | **MITU Skillologies** *Sep 2023 – Present | Pune, India*

- Built the reference GenAI systems behind the company's programmes: LangChain and LangGraph **RAG and agent pipelines**, fine-tuning workflows, and MLOps setups with Docker, MLflow, DVC and CI.
- Primary technical SME for the GenAI and agentic AI programmes (LangGraph, RAG, MCP), delivered to 2,000+ engineering students and faculty across 30+ institutions.

OPEN SOURCE

caura-long-run-fleet | **caura-ai** PR #43 · Merged | Oct 2026

Configurable per-agent subprocess timeout for the fleet simulator of a platform for long-running AI agents, with input validation and docs (fixes #36).

langchain-mongodb | **langchain-ai** PR #487 · Open | PR #486 · Open

Two fixes for LangGraph's **MongoDBStore**: suffix wildcard matching and limit handling in **list_namespaces**, with offline tests (fixes #480); and index desync for unindexed items in **put** and **batch** (fixes #479).

PROJECTS

Conversational Legal Agent (RAG) Live Demo | Source Code

RAG agent for Indian law with cited answers over **~70K chunks from 1,000+ acts**, also exposed as an **MCP server**. Built a **78-question retrieval eval** that caught a broken BM25 tokenizer and 2,718 mislabelled chunks; fixed and contained both, and measured vector search beating hybrid RRF fusion (MRR@10 0.56 vs 0.47).

Typed Decision Models: Agent Router Benchmark Source Code

Compared **XGBoost**, fine-tuned **BERT**, a fine-tuned typed decision model (Laya) and a **Gemini** router inside a **LangGraph** workflow. XGBoost matched the fine-tuned model at **~300x smaller size**; both beat the LLM router by **20+ points on exact match** at far lower latency and cost; a hybrid cascade made results worse.

Agentic Research System Live Demo | Source Code

LangGraph multi-agent system (Researcher, Writer, Critic) with typed state and conditional edges that loops back to the Writer only while the Critic's score is below threshold.

Industrial Equipment Maintenance (MLOps) Live Demo | Source Code

XGBoost classifier with a **DVC** pipeline, **MLflow** tracking, a **FastAPI** service in **Docker** and **GitHub Actions** CI.

SKILLS

Generative AI	RAG · Agentic AI · LLMs · LangGraph · LangChain · MCP · Gemini API · OpenAI API (incl. Azure OpenAI)
Retrieval & Embeddings	ChromaDB · Hybrid Search (BM25 + RRF) · Gemini Embeddings · Semantic Chunking · Hugging Face
Backend & Data	FastAPI · Pydantic · asyncio · DuckDB · MongoDB · Streamlit
ML / Deep Learning	XGBoost · Scikit-learn · PyTorch · Fine-tuning · NLP
MLOps & DevOps	Docker · GitHub Actions (CI/CD) · MLflow · DVC · pytest · Nginx
Languages	Python · SQL

EDUCATION & CERTIFICATIONS

PG Diploma in Artificial Intelligence — CDAC ACTS, Pune *2023*

B.Tech – Civil Engineering (CGPA: 7.78) — Savitribai Phule Pune University *2021*

Certificates: Generative AI and LLMs, IIM Bangalore (SWAYAM) · Supervised Machine Learning, DeepLearning.AI / Stanford